

# BasList vs 2.0.0

## for BasTools 1.0.0

Lists a BBC BASIC program file in Acorn or R.T. Russell (Z80) format with optional formatting and PrettyPrint syntax colouring.

## SYNTAX

```
BasList [/file=]filename ([line] | [line] [line]) | [line,line])
      [Options] ([IF ...] | [IFX ...] | [LIST ...])
```

```
BasList [/file=]filename [/V] [/notbasicv] [/addnumbers] [/align]
      [/indent[=(loops | defs | all)]] [/nonumbers] [/noformat]
      [/bare] [/splitlines] [/pause] [/prettyprint] [/dark | /light]
      [(/clear | /cls)] [/savedefaults]
```

BasList /resetdefaults - all other parameters are ignored

BasList /? or /h - help

## Options

/V Allow BASIC V keywords and assembler. See Compatibility below.

/addnumbers Show line numbers (starting at 10, in tens) when none in the program (Z80 only)

/align Right-align line numbers

Without /align (line numbers displayed with one following space):

```
10 REM An example program
100 MODE 7
1000 DEFPROCmain
```

With /align (line numbers ranged right and one following space):

```
10 REM An example program
100 MODE 7
1000 DEFPROCmain
```

/indent=loops Indent listing of loops (including WHILE), CASE and multiline IF structures.

Approximately equivalent to LISTO7 in BASIC IV, LISTO 2 in BASIC V.

/indent=defs Indents the body of PROCs and FNs

/indent Equivalent of /indent=all

/nonumbers Omits line numbers

/prettyprint Adds additional spaces and syntax colouring

/cls Clear screen. Clear console window before listing

|                |                                                                                                                                                                                                                                                                           |
|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| /dark          | Dark mode – black background (default). Best used with /cls.                                                                                                                                                                                                              |
| /light         | Light mode – white background (best used with /cls)                                                                                                                                                                                                                       |
| /noformat      | Preserves original program lines without spaces after line numbers. Cancels /prettyprint and /splitlines, but /indent is respected.)                                                                                                                                      |
|                | <pre> 10REM An example program 100MODE 7 1000DEFPROCmain </pre>                                                                                                                                                                                                           |
| /splitlines    | Splits lines at colons, THEN and ELSE.                                                                                                                                                                                                                                    |
|                | <pre> 70  FOR I% = 1 TO 4:            PRINT w\$(I%):            x% = x% + part(I%) 80      y% = win% - x%:            NEXT 90  ENDPROC </pre>                                                                                                                             |
| /bare          | Omits additional messages. If <i>not</i> specified, BasList displays an enhanced header with format information before the listing.                                                                                                                                       |
| /savedefaults  | After all switches have been processed, the current options (except filename, line numbers, directives etc.) will be saved. On next usage, these values will be loaded before any input is processed. To override a saved default, use /-<br>E.g. /-align cancels /align. |
| /resetdefaults | Returns saved defaults to application defaults. Should be used on its own.                                                                                                                                                                                                |

## EXAMPLES

|                              |                                                 |
|------------------------------|-------------------------------------------------|
| BasList Program ,200         | – List up to line 200                           |
| BasList Program 1000,        | – List from line 1000                           |
| BasList Program 200,1000     | – List from line 200 to 1000                    |
| BasList Program 200 1000     | – List from line 200 to 1000                    |
| BasList Program IF PRINTTAB  | – List only lines containing the specified text |
| BasList Program IFX          | – IF, but respecting spaces and case            |
| BasList Program LIST PROCinp | – Displays DEFPROCinp                           |
| BasList Program LIST FNadd   | – Displays DEFFNadd                             |

Note that there may be multiple PROCs and FNs specified, e.g. LIST PROCinp FNadd

## NOTES

- Options may be specified in any order, must start with a space and / and can be abbreviated. Note that **all** matching options apply. For example, /p activates both /pause and /prettyprint, /d both /dark and /debug.
- Parameters containing spaces must be enclosed by double quotes.
- /?, /h, or when no option is specified – display syntax help.
- Any IF, IFX or LIST clause *must* be at the end. Multiple matches may be entered and BasList will list any line containing at least one of them (in the case of IF/IFX) or any procedure or function specified.
- If you are producing a listing to be \*EXEC'd into a BBC Micro, use /bare /noformat to ensure a clean listing.

## FURTHER INFORMATION

### 1. Compatibility

BasList automatically determines whether the file is in Acorn or Z80 format. In Acorn mode it is compatible with:

- BBC BASIC II to Basic IV
- the version of BASIC issued with the Master Compact, sometimes known as BASIC 6
- BASIC V as issued with RiscOS

In Z80 mode it is compatible with:

- Z80 BASIC format or any BASIC version by R.T. Russell

In Acorn mode, BasList will set the 'Basic V' flag automatically when it encounters the extended keywords of Basic V, but we recommend using the /V switch as appropriate to ensure it will recognise ARM assembly language instead of 6502, even if the assembler comes before any Basic V tokens.

In Z80 mode, it is compatible with BBC BASIC on Zilog Z80 and Intel 8086+ processors, such as BASIC for CP/M machines, Amstrad Colour Computers (CPC464, CPC664 and CPC6128), Tatung Einstein and Cambridge Z88, as well as BBC BASIC (86) for IBM compatibles. Z80 assembler is supported.

### 2. Filenames

If the filename includes spaces, it must be enclosed by double quote characters ("), e.g.

```
BasList "C:\Users\My Documents\Basic Programs\Example"
```

If the filename is numeric it will be mistaken for a line number unless preceded by /file=

### 3. Line Numbers

By default, BasList displays the entire program.

As with the BBC Micro, you may limit the lines listed using:

- `number` – e.g. `100` – lists one line
- `number,` – e.g. `100,` – lists from line 100
- `, number` – e.g. `,1000` – lists up to line 1000

Note that the comma must be last or first character with no spaces.

- A range – e.g. `100,200` – May not contain spaces, whether quoted or not.
- A range – e.g. `100 200`.

If you specify the numbers in reverse, BasList will swap them round.

### 4. Options

Also called ‘switches’, each option must be preceded by a space and slash (/).

They may be in upper, lower or mixed case.

The order in which they appear is not significant.

Options may be shortened; you need only enter as much as is needed to distinguish them from other options. Thus `/noformat` may be no shorter than `/nof` if it is to be distinguished from `/nonumbers`. (`/n` activates **both** options.)

A non-existent option (preceded by `/` or `/-`) will result in a message (unless preceded on the line by `/bare`) but will otherwise be ignored.

### 5. Directives: IF, IFX, LIST

Directives *must follow* options / switches.

#### IF, IFX

Approximately equivalent to `LIST IF` in BASIC IV and above.

**IF** performs a case-insensitive match that is not restricted to whole words.

**IFX** (IF eXact) performs case-sensitive comparison and respects spaces.

You may use *either* IF *or* IFX but not both, and cannot combine with List.

The switches are not case sensitive, i.e. IF, if, If and iF are equivalent.

IF/IFX is followed by a list of conditions. For example,

```
BasList Program IF print colour red
```

will cause any line containing ‘print’ *or* ‘colour’ *or* ‘red’ to be listed.

To include a literal quote ("), escape it like this `\"` or double it like this `""`.

## IF

The comparison is not case sensitive. It also ignores spaces in the program being listed unless they are between quotes (literal strings) or in a REM statement.

IF PRINTTAB will show lines containing both PRINTTAB and PRINT TAB.

IF printer will show a line containing PRINT ERR, because PRINTERR contains 'printer' when case is ignored.

To match a literal space (e.g. in a REM or string), enclose the search term in double quotes.

**Detail:** When an IF clause is specified, BasList makes an internal copy of each program line, removing all spaces except those in quoted strings or in REM statements. Each match string is checked against the copy to see if it is contained in the line using case-insensitive comparison. It lists the line if any of them match. This has the advantage that it will, for example, find both PRINT TAB( and PRINTTAB(, but has the disadvantage that there is no way to limit the match to whole words.

## IFX

Case-sensitive comparisons are made against the *original* program line, respecting spaces.

```
BasList program IFX print
```

will list lines with 'printer', 'PROCrubandprint' but not PRINT.

If you are attempting to match a literal space, enclose the search string in double quotes.

IFX "PRINT ERROR" will find that literal text where it appears in quotes or REMs in the program, but

IFX "PRINT ERR" will match PRINT ERROR but not PRINTERR.

You cannot use keyword abbreviations. On a Master 128, LISTIF P. will find 'PRINT'; BasList works purely at the textual level.

Also consider saving the listing (see below), opening in a text editor and using Find.

For a syntax-aware search, see Advanced Search in BasViewer.

## LIST

This directive displays the definitions of procedures and functions. Listing is terminated when the next significant line (skipping blank lines, colons, DATA and REMs) is a DEF.

Note that procedure names must be given in their entirety, e.g. FNadd, PROCinp, *without* any brackets and arguments.

List is not case sensitive (fnadd, PROCINP will both match).

Line number range is ignored if specified.

## 6. Using the listings

To display the listing in ‘paged mode’, use the `/pause` option. The listing will stop at the bottom of the screen and wait for a keypress. Pressing Space will advance to the next screenful. Return will advance one line. Escape breaks off the listing. (Other than Escape, these keypresses have been chosen for compatibility with MORE<sup>1</sup>.)

To save the text to a file, add `>filename` to the end. E.g.

```
BasList program >"file.txt"
```

This will save (spool) the listing to a file named `file.txt` in the current directory, *instead of* displaying it on screen. Enclose the file specification in double quotes if it includes spaces.

To append several listings, use `>>filename`. Example:

```
BasList program 10,100 >file.txt
```

```
BasList program 1000,1200 >>file.txt
```

```
BasList program 3000,3100 >>file.txt
```

Note that the **first** command has **one** chevron, in order to create the text file, and subsequent ones have a **double** chevron to append to it.

Saving a text file in order to \*EXEC into a BBC Micro or copy and paste into BBC BASIC for Windows (for example) is a good way to convert between the Acorn and R.T. Russell formats.

## Version Information

1.0.0 First release

1.0.1 Bugs fixed:

- PrettyPrint added space after TO in the keyword TOP
- Now respects increased line number range for BASIC V

1.0.2 Enhancement:

- Removed the limitation that the filename must be the first parameter. It may now appear anywhere in the list of parameters (though IF or IFX must be last). The first parameter not preceded by `/` or `-` and not a line number will be treated as a file.
- Added `/file=` or `/f=` option for cases when filename is numeric and would otherwise be seen as a line number.

1.0.3 Installer adds batch files for Shell:sendto folder

1.0.4 Upgrade to .Net 8.0. Dark and light mode. PrettyPrint enhancements and bug fixes.

1.1.0 Major improvements to code analysis, and colouring and spacing with PrettyPrint.

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<sup>1</sup> Note that the DOS command MORE has the disadvantage of suppressing PrettyPrint colours.

IFX now compatible with PrettyPrint.

Addition of LIST directive.

ARM and 6502/65C12 assembler recognised (not Z80).

BreakApart option added (now SplitLines)

2.0.0 Correct indentation of each segment when /splitlines used

/savedefaults and /resetdefaults added. As a consequence, - instead of / removed

Enhanced recognition and formatting of BASIC syntax

Z80 assembler recognised

Full integration with the BasTools toolchain

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